TECHCIRCUIT HACKATHON | ROUND 2 UPDATED SUBMISSION

BrandGuard AI

Intelligent Brand Governance Platform

PROBLEM STATEMENT PROVIDER
Inkloom AI

TEAM NAME
ChipSmiths

TEAM MEMBERS AND PROPOSED FOCUS

1	**Syeed Hussain S H**	Coordination and integration
2	**U Tamilarasu**	Brand rules and test cases
3	**Shyam G R**	Python backend and checking logic
4	**Sandep S**	Interface, testing and documentation

Updated project report: the Round 1 proposal, revised with the Round 2 prototype.

01 OVERVIEW

Project Overview and Problem Statement

Content drifts from approved guidelines when people and AI tools write it

Brand content created by people and AI tools often drifts from approved guidelines. It can contain inconsistent messaging, outdated facts, unsupported promotional claims, incorrect terminology, and deviations from brand standards. BrandGuard AI is a proposed text-based platform that checks submitted content against approved brand rules and verified reference facts. Each finding is linked to supporting evidence where available. Claims that cannot be verified are marked “Needs Verification” rather than judged automatically. The platform proposes context-preserving corrections and routes flagged items to human reviewers, who make the final approval decision.

Problem statement. Brand standards drift across content, and manual review does not scale.

CORE PRINCIPLES

Evidence-linked

Findings tied to approved references

Uncertainty-aware

Unverifiable claims flagged for review

Correction suggestions

Context-preserving rewrites to consider

Human review

People make the final decision

Understanding the Problem

Brand standards drift across content, and manual review does not scale.

1

Inconsistent messaging

Tone, product names and key claims vary across content and channels.

2

Outdated information

Facts, dates or offers stay in content after they have changed.

3

Unsupported claims

Promotional statements appear without verifiable evidence.

4

Incorrect terminology

Product terms and technical names are used inconsistently or wrongly.

5

Brand deviations

Content departs from approved style, tone and formatting rules.

Problem this project solves: detect these issues in submitted text, link findings to approved evidence where available, and route uncertain cases to human review.

Existing Problems vs Current Solutions vs Our Solution

How each challenge is handled today, and how BrandGuard AI improves the process

3.1 Existing Problems vs. Current Solutions vs. Our Solution

The table below shows the challenges organisations face, how they are currently addressed, and how BfandGuard AI improves the process.

Existing Problem	Current Solution Used	Limitation of Current Solution	Our Solution – BrandGuard AI
Inconsistent brand messaging	Manual proofreading and brand guidelines	Depends on the reviewer's knowledge and attention.	Checks content against approved terminology, tone, and messaging rules.
Outdated product information	Manual fact-checking and reference documents	Requires reviewers to search for the latest approved information.	Compares statements with stored verified facts and flags potential conflicts.
Unsupported promotional claims	Manual verification and separate research tools	Supporting evidence may not be directly connected to each finding.	Links supported findings to relevant factual references and flags uncertain claims.
Incorrect terminology	Spreadsheets, style guides, and keyword-based tools	May miss context-dependent or semantically related issues.	Combines predefined rules with semantic analysis.
Unclear error explanations	Basic content-checking tools	Warnings may not explain the relevant organisational requirement.	Explains each finding and identifies the applicable guideline or reference where available.
Time-consuming review	Manual checking and separate tools	Repeated work across documents and teams.	Integrates analysis, correction suggestions, and review into one workflow.
Repeated compliance mistakes	Manual records and periodic reports	Difficult to identify recurring patterns consistently.	Uses governance analytics to highlight frequent violations and possible guideline gaps.

WHAT CHANGES

Manual proofreading depends on each reviewer's knowledge and attention. BrandGuard AI checks every submission against approved terminology, tone and messaging rules.

Reviewers no longer search for the latest approved facts by hand. Statements are compared with stored verified facts, and possible conflicts are flagged.

Each finding names the applicable guideline or reference where one exists, and governance analytics show repeated violations and possible guideline gaps.

04 STAKEHOLDERS

Users Affected

Five groups face the same review burden in different ways

Content creators

CHALLENGE

Unclear which claims and terms are approved.

BRANDGUARD HELPS

Instant checks with explanations before submission.

Marketing teams

CHALLENGE

Campaigns need fast review across many assets.

BRANDGUARD HELPS

Prioritised findings reduce manual checking time.

Brand managers

CHALLENGE

Guidelines are scattered and rarely applied consistently.

BRANDGUARD HELPS

A central rule base with review records.

External agencies

CHALLENGE

Work must match standards they have not seen before.

BRANDGUARD HELPS

Shared rules let agencies self-check drafts.

Organisations using AI

CHALLENGE

AI text can read well yet be unsupported or outdated.

BRANDGUARD HELPS

Evidence links and “Needs Verification” flags.

SHARED OUTCOME

One consistent, evidence-linked review workflow that every group can follow, with people making the final call.

Proposed Solution

A text-based checker that flags issues, shows evidence and suggests corrections

1. **Check against brand guidelines**
   Compares submitted text with approved brand rules.

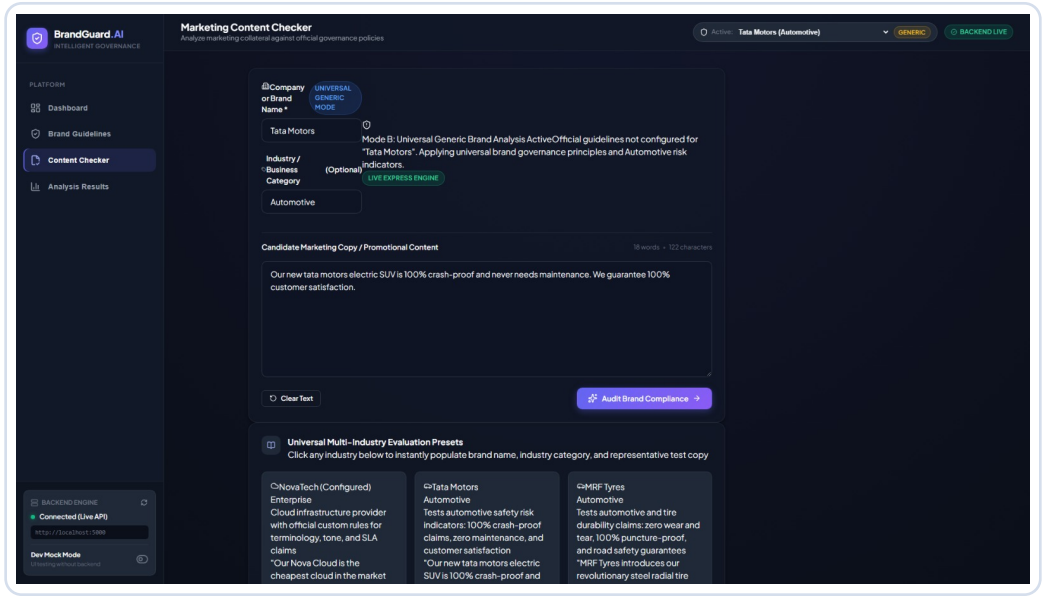
2. **Verify against facts**
   Matches claims with approved reference facts.

3. **Identify potential violations**
   Flags deviations, outdated facts and unsupported claims.

4. **Retrieve supporting references**
   Links each finding to the evidence used.

5. **Suggest corrections**
   Proposes edits that keep the original meaning.

6. **Route uncertain cases to reviewers**
   Items without enough evidence get “Needs Verification” status.



Content Checker in use. Submitted copy is audited against the active brand profile. Each issue shows its severity, category and a suggested replacement.

Complete Product Workflow

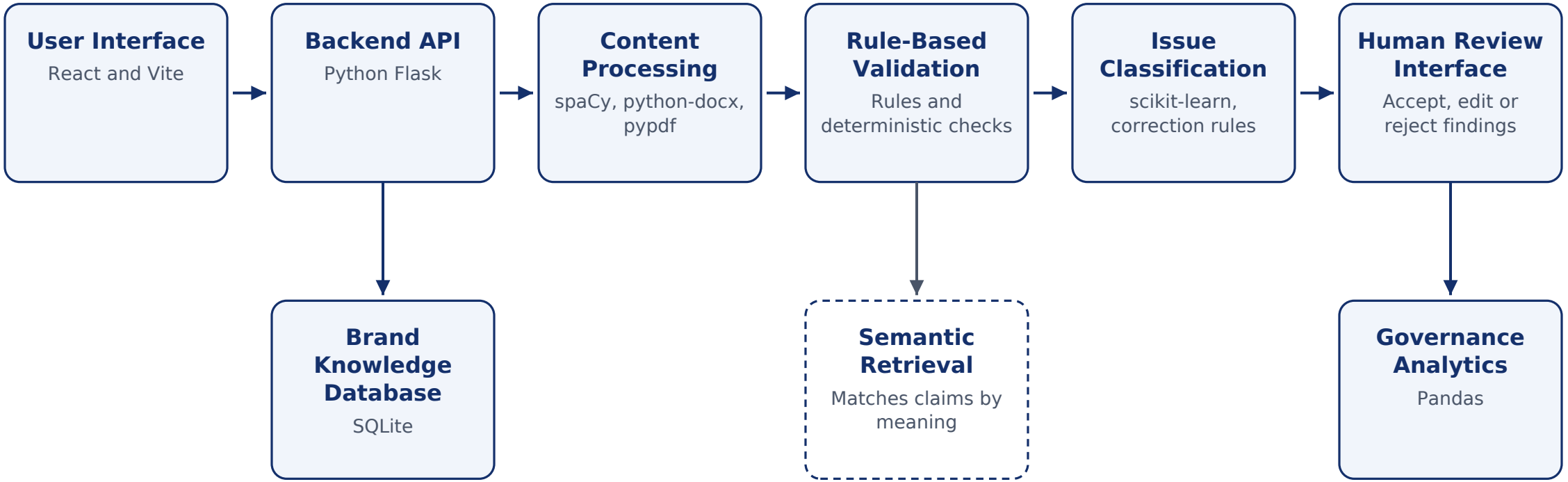
Eight stages from submission to records and analytics



Routing rule: where evidence is missing or insufficient, the item is marked “Needs Verification” and sent to human review.

System Architecture

Planned components and data flow for the BrandGuard AI prototype



Submitted text moves left to right through the checking pipeline. The brand knowledge database supports the API, and governance analytics draw on stored review records.

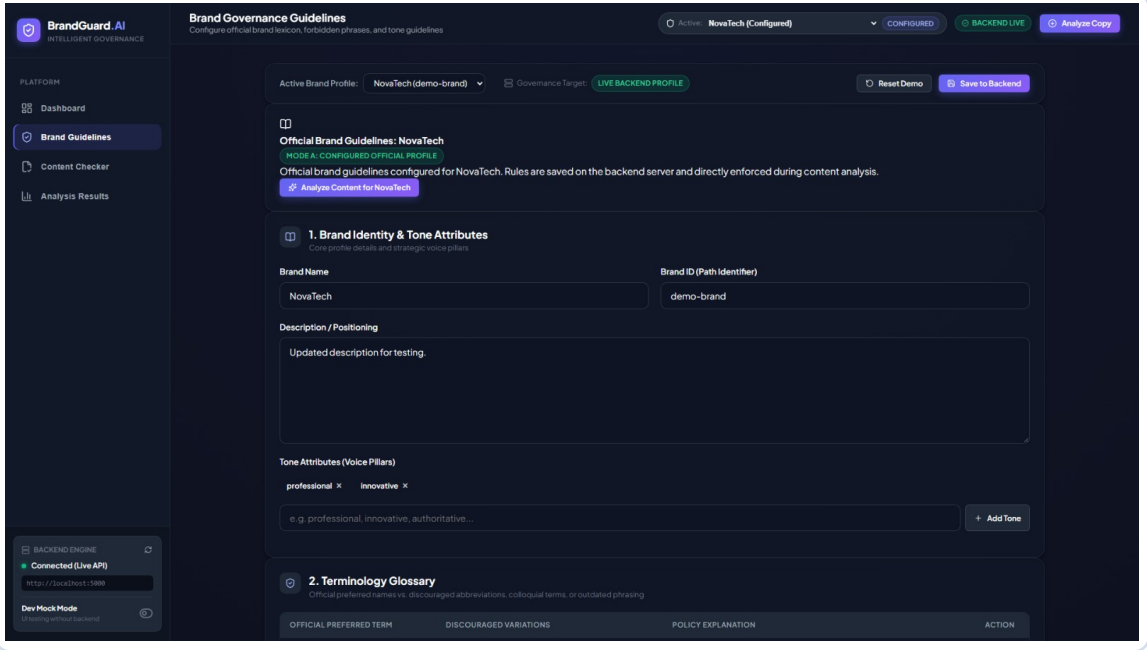
Main Features

Core prototype features are separated from those added only if time permits

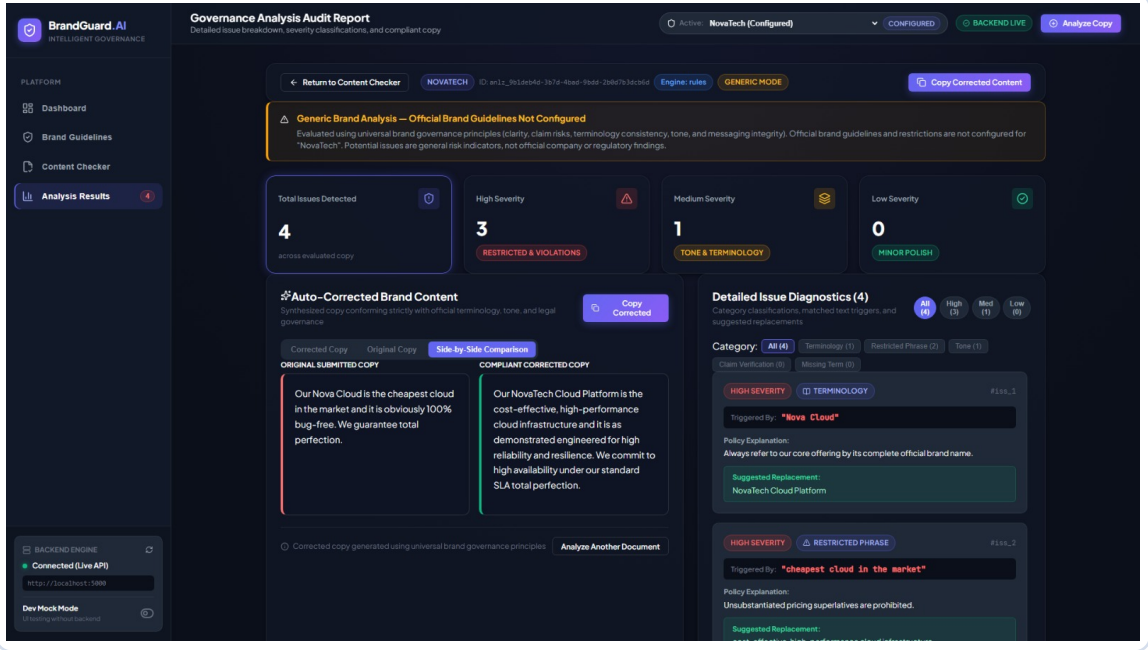
Module	Core function	Scope	Status in Round 2
1. Brand Knowledge Management	Store approved rules, terms and reference facts	Core prototype	**Implemented**
2. Content Ingestion and Processing	Accept text input and split it into analysable units	Core prototype	**Partial: text input only**
3. Brand Compliance Analysis	Apply rule-based checks for terminology, claims and standards	Core prototype	**Implemented**
4. Evidence-Linked Verification Engine	Link findings to references; semantic matching as an extension	Core, semantic if time permits	**Partial**
5. Explainable Correction Assistance	Suggest context-preserving rewrites with a stated reason	Core prototype	**Implemented**
6. Human Review and Approval	Reviewers accept, edit or reject each finding	Core prototype	**Not yet in prototype**
7. Governance Analytics	Trends in recurring violations and review outcomes	If time permits	**Not yet in prototype**

09 PROTOTYPE

Implemented Features and Working Website



Brand Governance Guidelines. Terminology glossary, tone attributes and saved brand profile.



Governance Analysis Audit Report. Issues grouped by severity, with side-by-side corrected copy.

COMPLETED FEATURES

- Brand profile setup: positioning, tone pillars and glossary
- Content checks run against the active brand profile
- Issues grouped by High, Medium and Low severity
- Suggested replacement and policy reason for each finding
- Auto-corrected copy with side-by-side comparison
- Generic Mode for brands without configured guidelines

Technology Stack and Implementation Details

Planned choices for a text-first prototype, with the interface connected to a live backend API

FRONTEND

React and Vite

Fast builds and a component-based interface

BACKEND

Python Flask

Lightweight API suited to a student prototype

DATABASE

SQLite

Zero-setup storage for rules, documents and reviews

LANGUAGE PROCESSING

spaCy

Sentence splitting and term matching

SEMANTIC SIMILARITY (STRETCH)

sentence-transformers

Matches claims to references by meaning

CLASSIFICATION

scikit-learn

Simple issue categorisation

DOCUMENT PROCESSING

python-docx, pypdf

Reads Word and PDF submissions (planned)

ANALYTICS (VISUAL OPTIONAL)

Pandas, Pillow, OpenCV

Recurring violation summaries; basic image checks if time permits

Implementation details. The interface connects to a live backend API at localhost:5000, and a Dev Mock Mode allows UI testing without the backend. Testing uses pytest with a hand-checked sample dataset.

Most challenging part. Linking each detected issue to relevant approved evidence and handling claims that cannot be verified. Approach: rule-based validation first, semantic retrieval second, and human review for every unresolved case.

Changes from Round 1 and Known Limitations

Changes from Round 1

- Round 1 covered the problem definition, architecture and planning only, with no implementation.
- Round 2 delivers a working prototype with a Brand Guidelines page for terminology, tone and saved profiles.
- The Content Checker sends submitted copy to the backend and returns issues with severity and category.
- Each finding shows the triggering text, a policy explanation and a suggested replacement.
- Auto-corrected copy and a side-by-side comparison are generated from the findings.
- Brands without configured guidelines use Universal Generic Mode, which applies general brand principles.

Known Limitations

- Checks are rule-based; semantic matching of claims to references is not yet included.
- Generic Mode findings are general risk indicators, not official company or regulatory findings.
- Evidence linking and “Needs Verification” routing are not yet fully shown in the prototype.
- Input is text pasted into the Content Checker; Word and PDF document processing is planned.
- Human review actions and governance analytics are not yet demonstrated.
- Impact and evaluation results are pending and have not yet been measured.

Planned Round 3 Improvements

Three improvements focused on brand relevance, claim verification and the approval workflow

1

Improve brand relevance detection

Evaluate content in its industry and product context, reducing incorrect brand mismatch warnings and preserving relevant product names.

2

Strengthen claim verification

Distinguish unsupported claims from verified information, and give evidence or a clear explanation for every claim that needs verification.

3

Brand guideline management and approval workflow

Let users define brand rules, review flagged content, and approve content before publication.

EVALUATION PLAN (RESULTS PENDING)

Compliant content should produce no findings.

Known violations should be detected and explained.

Outdated facts and unsupported claims should be marked Needs Verification.